

Django Tutorial #13 - Templates Basics & Variables

Quick Revision Notes | Django Framework | 5-Minute Quick Scan

1. PROJECT & APP SETUP QUICK STEPS

Step-by-Step Configuration:

1. **Project Create karein:** `django-admin startproject myproject5`
2. **App Create karein:** `python3 manage.py startapp blog`
3. **Settings.py mein App Register karein:** `INSTALLED_APPS` list ke andar `'blog'` add karein.
4. **Templates Directory Setup:** `TEMPLATES` settings mein `DIRS`: `[BASE_DIR / 'templates']` set karein taaki Django templates ko detect kar sake.

2. DJANGO COMMENTS TYPES

Templates mein code hide karne ke 3 tareeqe hain:

- **HTML Comment** `<!-- comment -->`: Yeh browser ke **Source Code** mein dikhta hai (Not recommended).
- **Django Comment (Multi-line)** `{% comment %} ... {% endcomment %}`: Yeh **recommended** hai, UI aur Source Code dono jagah hide rehta hai.
- **Single-Line Comment** `{# comment #}`: Yeh bhi completely hidden rehta hai.

3. VIEWS & CONTEXT PASSING

Views (`views.py`) mein dictionary ke roop mein data pass kiya jata hai:

```
def home(request):  
    context = {'name': 'Mohit', 'age': 30, 'skills': ['Python', 'Django']}  
    return render(request, 'blog/home.html', context)
```

4. ACCESSING VARIABLES IN TEMPLATES

Variables ko `{{ variable_name }}` syntax se HTML mein print karte hain:

- **Single Variable:** `Name` `{{ name }}` & `Age` `{{ age }}`
- **List Elements:** Full list `{{ skills }}` ya specific index `{{ skills.0 }}` (Python)
- **Object Attributes:** Class attributes `{{ user.name }}`, `{{ user.age }}`
- **Nested Variables:** Dictionary inside dictionary `{{ blog.author.name }}`

5. ADVANCED VARIABLE FILTERS & MODIFIERS

- **Safe Filter (`|safe`):** Agar variable mein HTML tags (`<b>`) hain aur aap chahte hain ki browser use render kare (raw text print na ho), toh `{{ blog.content | safe }}` use karein.
- **Default Value (`|default`):** Agar variable empty/false hai toh default text dikhane ke liye `{{ empty_val | default:"No Value Provided" }}` use karein.
- **Default if None (`|default_if_none`):** Jab value explicitly None ho tabhi fallback ke liye.

6. QUICK REVISION / KEY TAKEAWAYS

- **Setup Checklist:** App registration `INSTALLED_APPS` mein aur template path `TEMPLATES DIRS` mein define karna zaroori hai.
- **Rendering:** `render(request, 'template_name.html', context)` function use hota hai.
- **Syntax Reminders:** Variables ke liye `{{ ... }}` aur tags/logic ke liye `{% ... %}` syntax use hota hai.